

# Master Proof Compendium

STAT3006/7305 — Statistical Learning

A comprehensive, lecture-by-lecture collection of every proof  
and major derivation in the course material

Compiled from the lecture slides

University of Queensland

Each result is stated with a plain-language framing followed by the formal argument,  
mirroring how it appears in the corresponding lecture deck.

# Contents

|                                                                          |           |
|--------------------------------------------------------------------------|-----------|
| <b>How this document is organised</b>                                    | <b>2</b>  |
| <b>1 Supervised Learning: <math>k</math>-Nearest Neighbours</b>          | <b>3</b>  |
| 1.1 The Bayes classifier is optimal for 0–1 error . . . . .              | 3         |
| 1.2 Cover–Hart Theorem: the asymptotic error of 1-NN . . . . .           | 3         |
| 1.3 Stone’s Theorem: universal consistency of $k$ -NN . . . . .          | 4         |
| <b>2 Supervised Learning: Decision Trees</b>                             | <b>4</b>  |
| 2.1 The optimal leaf prediction is the sample mean . . . . .             | 4         |
| 2.2 Probabilistic meaning of the Gini index . . . . .                    | 5         |
| 2.3 Consistency of regression trees . . . . .                            | 5         |
| <b>3 Linear Regression: Optimisation Perspective</b>                     | <b>6</b>  |
| 3.1 The normal equations . . . . .                                       | 6         |
| 3.2 Closed form for ridge regression . . . . .                           | 6         |
| <b>4 Linear Regression: Probabilistic Perspective</b>                    | <b>7</b>  |
| 4.1 Gaussian-noise MLE equals least squares . . . . .                    | 7         |
| 4.2 The Gaussian posterior (Bayesian linear regression) . . . . .        | 8         |
| 4.3 MAP estimation recovers ridge regression . . . . .                   | 8         |
| 4.4 Posterior predictive distribution . . . . .                          | 9         |
| <b>5 Logistic Regression</b>                                             | <b>9</b>  |
| 5.1 Cross-entropy loss from maximum likelihood . . . . .                 | 9         |
| 5.2 The compact logistic loss . . . . .                                  | 9         |
| 5.3 From binary to multiclass: the softmax . . . . .                     | 10        |
| 5.4 Cross-entropy decomposes into entropy plus KL . . . . .              | 10        |
| 5.5 Logistic regression has a linear decision boundary . . . . .         | 10        |
| 5.6 Poisson regression negative log-likelihood . . . . .                 | 11        |
| <b>6 Loss Functions: Decision-Theoretic Foundations</b>                  | <b>11</b> |
| 6.1 Pointwise characterisation of the Bayes-optimal predictor . . . . .  | 11        |
| 6.2 Squared loss targets the conditional mean . . . . .                  | 11        |
| 6.3 Absolute loss targets the conditional median . . . . .               | 12        |
| 6.4 Negative log-likelihood is a proper loss . . . . .                   | 12        |
| 6.5 0–1 loss: Bayes classifier and impropriety . . . . .                 | 12        |
| 6.6 Cross-entropy and logistic losses are strictly proper . . . . .      | 13        |
| <b>7 Loss Functions: Regression and Classification</b>                   | <b>13</b> |
| 7.1 Noise model determines the regression loss . . . . .                 | 13        |
| 7.2 Distance to the decision boundary and the margin . . . . .           | 14        |
| <b>8 Evaluating Performance</b>                                          | <b>14</b> |
| 8.1 Hold-out validation is unbiased . . . . .                            | 14        |
| 8.2 ROC-AUC equals a pairwise ranking probability . . . . .              | 14        |
| <b>9 Model Complexity and Generalisation: Classical View</b>             | <b>15</b> |
| 9.1 The bias–variance decomposition of an estimator . . . . .            | 15        |
| 9.2 Bias–variance decomposition of the generalisation error . . . . .    | 15        |
| 9.3 Heuristic links: training error and the generalisation gap . . . . . | 16        |

|                                                                           |           |
|---------------------------------------------------------------------------|-----------|
| <b>10 Optimisation</b>                                                    | <b>16</b> |
| 10.1 Adam's bias correction . . . . .                                     | 16        |
| <b>11 Regularisation</b>                                                  | <b>17</b> |
| 11.1 Bias and variance of the ridge estimator . . . . .                   | 17        |
| 11.2 Variance of a linear combination . . . . .                           | 17        |
| <b>12 Kernel Methods: Kernel Ridge Regression</b>                         | <b>18</b> |
| 12.1 Kernelising ridge regression via the push-through identity . . . . . | 18        |
| 12.2 When is a kernel a valid inner product? PSD examples . . . . .       | 18        |
| 12.3 The Representer Theorem (finite-dimensional case) . . . . .          | 18        |
| <b>13 Kernel Methods: RKHS</b>                                            | <b>19</b> |
| 13.1 The Representer Theorem (RKHS form) . . . . .                        | 19        |
| 13.2 Bias and variance of kernel ridge regression . . . . .               | 20        |
| <b>14 Kernel Methods: Support Vector Regression &amp; Classification</b>  | <b>20</b> |
| 14.1 Weak duality . . . . .                                               | 20        |
| 14.2 The SVR dual . . . . .                                               | 20        |
| 14.3 The SVC (soft-margin SVM) dual . . . . .                             | 21        |
| <b>15 Kernel Methods: Gaussian Processes</b>                              | <b>22</b> |
| 15.1 Construction and existence . . . . .                                 | 22        |
| 15.2 The marginal distribution of the observations . . . . .              | 22        |
| 15.3 GP regression posterior by Gaussian conditioning . . . . .           | 22        |
| 15.4 Posterior variance is monotone decreasing in the data . . . . .      | 23        |
| <b>Closing note</b>                                                       | <b>23</b> |

## How this document is organised

This compendium gathers every proof and substantive derivation from the nineteen lecture decks. “Proof” is read broadly: it includes formally stated theorems with proofs (e.g. Cover–Hart, the Representer Theorem, tree consistency) *and* the step-by-step derivations the course treats as its working mathematics (e.g. the normal equations, the bias–variance decomposition, the SVR dual). Purely conceptual lectures (the *Introduction*, and most of the *Modern View* of complexity) contain no proofs and are noted but not reproduced.

Each entry follows the same pattern: a short **plain-language idea** of what is being shown and why it matters, then the formal statement and argument. Notation follows the slides:  $\langle \cdot, \cdot \rangle$  is an inner product,  $\eta(x) = \mathbb{P}(y = 1 \mid x)$  is the positive-class posterior,  $f^*$  is the Bayes-optimal predictor or true function, and a hat ( $\hat{\theta}$ ,  $\hat{f}$ ) denotes a learned estimate.

# 1 Supervised Learning: $k$ -Nearest Neighbours

## 1.1 The Bayes classifier is optimal for 0–1 error

### Plain-language idea

If we somehow knew the true probability  $\eta(x) = \mathbb{P}(y = 1 \mid x)$  at every point, the smartest possible guess is simply “predict the more likely class.” This rule, the *Bayes classifier*, cannot be beaten on average. It is the gold standard every real classifier is measured against.

**Definition 1.1** (Bayes classifier). For  $y \in \{0, 1\}$  and  $\eta(x) = \mathbb{P}(y = 1 \mid x)$ , the Bayes classifier is

$$f^*(x) = \begin{cases} 1 & \eta(x) \geq \frac{1}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

**Theorem 1.2** (Optimality of the Bayes classifier). For any classifier  $f : \mathbb{R}^d \rightarrow \{0, 1\}$  and every  $x$ ,  $\mathbb{P}(f^*(x) \neq y \mid x) \leq \mathbb{P}(f(x) \neq y \mid x)$ .

*Proof.* Conditioned on  $x$ , the error of a fixed prediction is

$$\mathbb{P}(f(x) \neq y \mid x) = \begin{cases} 1 - \eta(x) & f(x) = 1, \\ \eta(x) & f(x) = 0. \end{cases}$$

The classifier chooses, pointwise, between the two values  $\eta(x)$  and  $1 - \eta(x)$ . If  $\eta(x) \geq \frac{1}{2}$  then  $f^*(x) = 1$  incurs error  $1 - \eta(x) \leq \eta(x)$ ; if  $\eta(x) < \frac{1}{2}$  then  $f^*(x) = 0$  incurs error  $\eta(x) < 1 - \eta(x)$ . In both cases  $f^*$  attains the smaller of the two possible errors, so  $f^*(x) = \arg \min_{t \in \{0, 1\}} \mathbb{P}(f(x) \neq y \mid x)$ . Since this holds at every  $x$ , it holds in expectation, giving the global optimum. The minimum attainable conditional error is the *Bayes error*  $R^*(x) = \min\{\eta(x), 1 - \eta(x)\}$ , with overall value  $R^* = \mathbb{E}_x[R^*(x)]$ .  $\square$

## 1.2 Cover–Hart Theorem: the asymptotic error of 1-NN

### Plain-language idea

With unlimited data, the single nearest neighbour of a query point sits essentially on top of it, so its label is an independent coin-flip from the same distribution as the query’s own label. A 1-NN mistake is then “two draws of the same biased coin disagree.” Working this out shows 1-NN’s error is at most *twice* the unbeatable Bayes error — remarkably good for such a crude rule, but never optimal, because it adds its own sampling noise on top of the inherent noise.

### Approach

The argument is built up over several slides: (i) characterise the label of the nearest neighbour as  $n \rightarrow \infty$ ; (ii) compute the pointwise disagreement probability; (iii) rewrite it in terms of the Bayes error; (iv) integrate and bound. We follow the same four steps.

**Setup.** Training pairs  $(x_i, y_i)$  are i.i.d. with  $y \in \{0, 1\}$ ; a test pair  $(x, y)$  is drawn independently. Let  $(x_{(1)}, y_{(1)})$  be the nearest neighbour of  $x$ . Define  $R_{1\text{-NN}} = \lim_{n \rightarrow \infty} \mathbb{P}(f_{1\text{-NN}}(x) \neq y)$ .

**Step 1 — limiting behaviour of the neighbour.** As  $n \rightarrow \infty$ ,  $x_{(1)} \rightarrow x$ , so by continuity of  $\eta$ ,  $\eta(x_{(1)}) \rightarrow \eta(x)$ . The neighbour label satisfies  $\mathbb{P}(y_{(1)} = 1 \mid x_{(1)}) = \eta(x_{(1)}) \rightarrow \eta(x)$ .

Conditioned on  $x$ , the query label  $y$  and the neighbour label  $y_{(1)}$  become *independent* draws from  $\text{Bernoulli}(\eta(x))$ .

**Result/Derivation 1.3** (Pointwise 1-NN error).

$$R_{1\text{-NN}}(x) = \mathbb{P}(y \neq y_{(1)} \mid x) = 2\eta(x)(1 - \eta(x)).$$

*Proof.*

$$\begin{aligned} R_{1\text{-NN}}(x) &= \mathbb{P}(y = 1, y_{(1)} = 0 \mid x) + \mathbb{P}(y = 0, y_{(1)} = 1 \mid x) \\ &= \eta(x)(1 - \eta(x)) + (1 - \eta(x))\eta(x) = 2\eta(x)(1 - \eta(x)), \end{aligned}$$

using conditional independence  $\mathbb{P}(A, B \mid C) = \mathbb{P}(A \mid C)\mathbb{P}(B \mid C)$ . This is exactly twice the variance of a  $\text{Bernoulli}(\eta(x))$  variable.  $\square$

**Lemma 1.4** (Relating to the Bayes error).  $2\eta(x)(1 - \eta(x)) = 2R^*(x)(1 - R^*(x))$ , where  $R^*(x) = \min\{\eta(x), 1 - \eta(x)\}$ .

*Proof.* The map  $z \mapsto z(1 - z)$  is symmetric about  $z = \frac{1}{2}$ . If  $\eta(x) \leq \frac{1}{2}$  then  $R^*(x) = \eta(x)$  and the right side is  $2\eta(x)(1 - \eta(x))$ . If  $\eta(x) > \frac{1}{2}$  then  $R^*(x) = 1 - \eta(x)$  and the right side is  $2(1 - \eta(x))(1 - (1 - \eta(x))) = 2(1 - \eta(x))\eta(x)$ . Either way it equals  $2\eta(x)(1 - \eta(x))$ .  $\square$

**Theorem 1.5** (Cover–Hart, 1967).  $R_{1\text{-NN}} \leq 2R^*$ .

*Proof.* Integrating the pointwise error over  $x$  and using the lemma,

$$R_{1\text{-NN}} = \mathbb{E}_x[R_{1\text{-NN}}(x)] = \mathbb{E}_x[2R^*(x)(1 - R^*(x))].$$

Since  $R^*(x) \in [0, \frac{1}{2}]$  we have  $1 - R^*(x) \leq 1$ , hence  $R^*(x)(1 - R^*(x)) \leq R^*(x)$  and

$$R_{1\text{-NN}} \leq 2\mathbb{E}_x[R^*(x)] = 2R^*. \quad \square$$

### 1.3 Stone’s Theorem: universal consistency of $k$ -NN

#### Plain-language idea

A single neighbour can never average out local noise, so 1-NN stays a constant factor above optimal. If instead we let the number of neighbours  $k$  grow — but slowly, so the neighbourhood still shrinks to the query point — we average away the noise *and* keep locality, and the error converges exactly to the Bayes error.

**Theorem 1.6** (Stone, 1977 — stated). If  $k \rightarrow \infty$  and  $k/n \rightarrow 0$  as  $n \rightarrow \infty$ , then  $R_{k\text{-NN}} \rightarrow R^*$ . Under smoothness of  $\eta$ , the excess risk obeys

$$\underbrace{R_{k\text{-NN}} - R^*}_{\text{excess risk}} \lesssim \underbrace{\frac{1}{\sqrt{k}}}_{\text{estimation (variance)}} + \underbrace{\left(\frac{k}{n}\right)^{1/d}}_{\text{approximation (bias)}}.$$

The exponent  $1/d$  is the curse of dimensionality: to keep the error small,  $n$  must grow exponentially in  $d$ .

(Stated without proof in the lecture; the two conditions  $k \rightarrow \infty$  and  $k/n \rightarrow 0$  separately kill variance and bias.)

## 2 Supervised Learning: Decision Trees

### 2.1 The optimal leaf prediction is the sample mean

### Plain-language idea

Once a tree has carved the input space into regions, the only question left inside each region is: what single number should we predict? If we are scored by squared error, the best constant is just the average of the responses there.

**Result/Derivation 2.1** (Regression leaf minimiser). With the partition  $\{R_t\}$  fixed, the constant  $f_t$  minimising the sum of squared errors in region  $R_t$  is the regional mean:

$$f_t = \arg \min_c \sum_{i: x_i \in R_t} (y_i - c)^2 = \frac{1}{|R_t|} \sum_{i: x_i \in R_t} y_i.$$

*Proof.* Differentiate the convex objective in  $c$ :

$$\frac{d}{dc} \sum_{i: x_i \in R_t} (y_i - c)^2 = \sum_{i: x_i \in R_t} 2(c - y_i) = 2|R_t|c - 2 \sum_{i: x_i \in R_t} y_i.$$

Setting this to zero gives  $c = \frac{1}{|R_t|} \sum_{i: x_i \in R_t} y_i$ . □

## 2.2 Probabilistic meaning of the Gini index

### Plain-language idea

The Gini impurity of a node has a clean interpretation: draw two points at random from the node; Gini is the chance they belong to *different* classes. A pure node gives zero; a perfectly mixed node gives the most.

**Result/Derivation 2.2** (Gini as a collision probability). With class proportions  $\pi_t^c$  in node  $R_t$ , the Gini impurity cost  $\text{cost}(R_t) = 1 - \sum_c (\pi_t^c)^2$  equals the probability that two independent draws from the node have different classes.

*Proof.* Drawing two points independently (with replacement),

$$\mathbb{P}(\text{same class}) = \sum_{c=1}^C \mathbb{P}(y_1 = c, y_2 = c) = \sum_{c=1}^C (\pi_t^c)^2,$$

so  $\mathbb{P}(\text{different classes}) = 1 - \sum_{c=1}^C (\pi_t^c)^2$ , which is the Gini impurity. □

## 2.3 Consistency of regression trees

### Plain-language idea

Will a tree, given more and more data, eventually recover the true regression function? Yes — provided the leaves shrink (so each leaf sees a nearly-constant piece of the true function, killing bias) *and* each leaf collects ever more points (so averaging kills the noise variance). The proof splits the error into exactly these two pieces and sends each to zero.

### Approach

We (i) write the error at a fixed test point, (ii) decompose it into conditional variance plus conditional squared bias given the random leaf, (iii) show the bias term vanishes from shrinking leaves and continuity, and (iv) show the variance term vanishes from growing leaf counts. Dominated convergence lifts the almost-sure limits to limits in expectation.

**Theorem 2.3** (Tree consistency). Assume  $y = f^*(x) + \varepsilon$  with  $\mathbb{E}[\varepsilon \mid x] = 0$ ,  $\text{Var}(\varepsilon \mid x) = \sigma^2$ ,  $f^*$  bounded and continuous, and a tree whose leaves satisfy  $\max_t \text{diam}(R_t^{(n)}) \xrightarrow{a.s.} 0$  and  $\min_t |R_t^{(n)}| \xrightarrow{a.s.} \infty$ . Then for fixed  $x_0$ ,  $\mathbb{E}[(f^{(n)}(x_0) - f^*(x_0))^2] \rightarrow 0$ .

*Proof.* Let  $R^{(n)}(x_0)$  be the (random) leaf containing  $x_0$  and  $f^{(n)}(x_0) = \frac{1}{|R^{(n)}(x_0)|} \sum_{i: x_i \in R^{(n)}(x_0)} y_i$ . Conditioning on the leaf,

$$\mathbb{E}[(f^{(n)}(x_0) - f^*(x_0))^2] = \mathbb{E}\left[\underbrace{\text{Var}(f^{(n)}(x_0) \mid R^{(n)}(x_0))}_{\text{variance}} + \underbrace{(\mathbb{E}[f^{(n)}(x_0) \mid R^{(n)}(x_0)] - f^*(x_0))^2}_{\text{bias}^2}\right].$$

*Bias.* Since  $\mathbb{E}[y_i \mid x_i] = f^*(x_i)$ ,

$$\mathbb{E}[f^{(n)}(x_0) \mid R^{(n)}(x_0)] = \frac{1}{|R^{(n)}(x_0)|} \sum_{i: x_i \in R^{(n)}(x_0)} f^*(x_i).$$

As  $\text{diam}(R^{(n)}(x_0)) \xrightarrow{a.s.} 0$  and  $f^*$  is continuous at  $x_0$ , this average converges a.s. to  $f^*(x_0)$ , so the squared bias  $\rightarrow 0$  a.s. Because  $f^*$  is bounded, dominated convergence gives  $\mathbb{E}[\text{bias}^2] \rightarrow 0$ .

*Variance.* Using conditional independence of the responses,

$$\text{Var}(f^{(n)}(x_0) \mid R^{(n)}(x_0)) = \frac{1}{|R^{(n)}(x_0)|^2} \sum_{i: x_i \in R^{(n)}(x_0)} \text{Var}(y_i \mid x_i) = \frac{\sigma^2}{|R^{(n)}(x_0)|}.$$

Since  $|R^{(n)}(x_0)| \xrightarrow{a.s.} \infty$ , this  $\rightarrow 0$  a.s., and being bounded by  $\sigma^2$ , dominated convergence gives  $\mathbb{E}[\text{variance}] \rightarrow 0$ . Adding the two limits gives the claim. (Fully grown trees with  $|R^{(n)}(x_0)| \not\rightarrow \infty$  violate the variance condition, which is why they overfit.)  $\square$

### 3 Linear Regression: Optimisation Perspective

#### 3.1 The normal equations

##### Plain-language idea

Least squares asks for the parameter that makes the straight-line predictions as close as possible to the data, measured by total squared error. Because that objective is a smooth bowl (convex), we just set its slope to zero. Doing so produces one tidy linear system — the *normal equations* — whose solution is the best fit.

**Result/Derivation 3.1** (Normal equations). The least-squares objective  $f(\theta) = \frac{1}{2} \sum_{i=1}^n (\langle x_i, \theta \rangle - y_i)^2 = \frac{1}{2} \|X\theta - y\|_2^2$  is minimised at any  $\hat{\theta}$  with

$$X^\top X \hat{\theta} = X^\top y.$$

*Proof.*  $f$  is convex, so a minimiser is any stationary point. The gradient is

$$\nabla f(\theta) = \frac{1}{2} \sum_{i=1}^n \frac{\partial}{\partial \theta} (\langle x_i, \theta \rangle - y_i)^2 = \sum_{i=1}^n (\langle x_i, \theta \rangle - y_i) x_i.$$

Setting  $\nabla f(\hat{\theta}) = 0$  gives  $\sum_i \langle x_i, \hat{\theta} \rangle x_i = \sum_i y_i x_i$ , which in matrix form is  $X^\top X \hat{\theta} = X^\top y$ .  $\square$

**Corollary 3.2** (OLS closed form). If  $X$  has linearly independent columns,  $X^\top X$  is invertible and  $\hat{\theta} = (X^\top X)^{-1} X^\top y$ . If  $X^\top X$  is singular (redundant features, or  $p > n$ ), infinitely many  $\theta$  achieve the same minimal residual.

#### 3.2 Closed form for ridge regression



#### Plain-language idea

Ridge adds a penalty on large coefficients to tame overfitting. The penalty also has a happy side effect: it makes the linear system always solvable, even when ordinary least squares cannot decide between many equally good fits.

**Result/Derivation 3.3** (Ridge solution). The minimiser of  $\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \|X\theta - y\|_2^2 + \lambda \|\theta\|_2^2$  satisfies the first-order condition  $(X^\top X + n\lambda I)\hat{\theta} = X^\top y$ , and for  $\lambda > 0$ ,

$$\hat{\theta} = (X^\top X + n\lambda I)^{-1} X^\top y.$$

*Proof.* Setting the gradient of  $\frac{1}{n} \|X\theta - y\|_2^2 + \lambda \|\theta\|_2^2$  to zero gives  $\frac{2}{n} X^\top (X\theta - y) + 2\lambda\theta = 0$ , i.e.  $(X^\top X + n\lambda I)\theta = X^\top y$ . For  $\lambda > 0$  the matrix  $X^\top X + n\lambda I$  is positive definite (its eigenvalues are those of  $X^\top X$ , which are  $\geq 0$ , shifted up by  $n\lambda > 0$ ), hence invertible.  $\square$

## 4 Linear Regression: Probabilistic Perspective

### 4.1 Gaussian-noise MLE equals least squares

#### Plain-language idea

Why squared error? One deep answer: if you believe the noise around your line is bell-shaped (Gaussian), then “the line that makes the data most probable” is *exactly* the least-squares line. The optimisation recipe and the statistical recipe coincide.

#### Approach

Write the likelihood as a product over independent points, plug in the Gaussian density, take logs to turn the product into a sum, and discard the  $\theta$ -independent constant. What remains is the sum of squared residuals.

**Result/Derivation 4.1** (MLE under Gaussian noise). With  $y_i = \langle \theta, x_i \rangle + \varepsilon_i$  and  $\varepsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$ ,

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(y \mid X; \theta) = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n (\langle \theta, x_i \rangle - y_i)^2.$$

*Proof.* Independence factorises the likelihood,  $p(y \mid X; \theta) = \prod_i p(y_i \mid x_i; \theta)$ , and the Gaussian model gives  $p(y_i \mid x_i; \theta) = \mathcal{N}(\langle \theta, x_i \rangle, \sigma^2)$ , so

$$p(y \mid X; \theta) = \left( \frac{1}{2\pi\sigma^2} \right)^{n/2} \exp \left( -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \langle \theta, x_i \rangle)^2 \right).$$

Taking logs (monotone, so the maximiser is unchanged),

$$\ln p(y \mid X; \theta) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (\langle \theta, x_i \rangle - y_i)^2.$$

The first term is constant in  $\theta$ , so maximising the log-likelihood is equivalent to minimising  $\sum_i (\langle \theta, x_i \rangle - y_i)^2$ .  $\square$

**Corollary 4.2** (Laplace noise gives the  $\ell_1$  loss). If instead  $\varepsilon_i \stackrel{iid}{\sim} \text{Laplace}(0, \sigma)$ , then  $p(y_i \mid x_i; \theta) \propto \exp(-\frac{1}{\sigma} |y_i - \theta^\top x_i|)$ , and maximising the likelihood is equivalent to minimising  $\frac{1}{n} \sum_i |y_i - \theta^\top x_i|$ , the least-absolute-deviation problem.

## 4.2 The Gaussian posterior (Bayesian linear regression)

### Plain-language idea

Instead of one best line, the Bayesian view keeps a whole cloud of plausible lines and updates that cloud as data arrives. With a Gaussian prior and Gaussian noise the updated cloud is *again* Gaussian — the algebra works out because everything in the exponent is quadratic, and completing the square reveals the new mean and covariance.

### Approach

Posterior  $\propto$  likelihood  $\times$  prior. Both contribute quadratics in  $\theta$  inside the exponent. Collect the quadratic and linear terms, then complete the square to read off a Gaussian with new precision and mean.

**Result/Derivation 4.3** (Posterior is Gaussian). With likelihood  $p(y | X, \theta) = \mathcal{N}(X\theta, \sigma^2 I)$  and prior  $p(\theta) = \mathcal{N}(\mu_0, \Sigma_0)$ , the posterior is  $p(\theta | X, y) = \mathcal{N}(\mu_N, \Sigma_N)$ , where

$$\Sigma_N^{-1} = \frac{1}{\sigma^2} X^\top X + \Sigma_0^{-1}, \quad \mu_N = \Sigma_N \left( \frac{1}{\sigma^2} X^\top y + \Sigma_0^{-1} \mu_0 \right).$$

*Proof.* Since the evidence  $p(y | X)$  does not depend on  $\theta$ ,  $p(\theta | X, y) \propto p(y | X, \theta)p(\theta)$ . Expanding the two exponents and dropping  $\theta$ -independent constants,

$$\begin{aligned} -\frac{1}{2\sigma^2} \|y - X\theta\|^2 &= -\frac{1}{2\sigma^2} (\theta^\top X^\top X \theta - 2\theta^\top X^\top y) + \text{const}, \\ -\frac{1}{2} (\theta - \mu_0)^\top \Sigma_0^{-1} (\theta - \mu_0) &= -\frac{1}{2} \theta^\top \Sigma_0^{-1} \theta + \theta^\top \Sigma_0^{-1} \mu_0 + \text{const}. \end{aligned}$$

Adding them, the combined exponent is

$$-\frac{1}{2} \theta^\top \underbrace{\left( \frac{1}{\sigma^2} X^\top X + \Sigma_0^{-1} \right)}_{\Sigma_N^{-1}} \theta + \theta^\top \underbrace{\left( \frac{1}{\sigma^2} X^\top y + \Sigma_0^{-1} \mu_0 \right)}_{\Sigma_N^{-1} \mu_N}.$$

Using the completing-the-square identity  $-\frac{1}{2} \theta^\top A \theta + \theta^\top A b = -\frac{1}{2} (\theta - b)^\top A (\theta - b) + \frac{1}{2} b^\top A b$  with  $A = \Sigma_N^{-1}$  and  $b = \mu_N$ , the exponent becomes  $-\frac{1}{2} (\theta - \mu_N)^\top \Sigma_N^{-1} (\theta - \mu_N) + \text{const}$ . This is the exponent of  $\mathcal{N}(\mu_N, \Sigma_N)$ ; the normalising constant is forced by integration to one.  $\square$

## 4.3 MAP estimation recovers ridge regression

### Plain-language idea

If we just want the single most probable line from the Bayesian cloud (the MAP estimate) and our prior says “coefficients are probably small,” the calculation collapses exactly to ridge regression. Ridge’s penalty is a Gaussian prior in disguise.

**Result/Derivation 4.4** (MAP = ridge). With prior  $\theta \sim \mathcal{N}(0, \Sigma_0)$ ,  $\Sigma_0 = \frac{\sigma^2}{n\lambda} I$ , the MAP estimate  $\theta_{\text{MAP}} = \arg \max_{\theta} [\log p(y | X, \theta) + \log p(\theta)]$  equals the ridge minimiser of  $\frac{1}{n} \|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2$ .

*Proof.* Up to constants,  $\log p(\theta) = -\frac{1}{2} \theta^\top \Sigma_0^{-1} \theta = -\frac{n\lambda}{2\sigma^2} \|\theta\|_2^2$  and  $\log p(y | X, \theta) = -\frac{1}{2\sigma^2} \|y - X\theta\|_2^2$ . Their sum is  $-\frac{1}{2\sigma^2} (\|y - X\theta\|_2^2 + n\lambda \|\theta\|_2^2)$ . Maximising this is the same as minimising  $\frac{1}{n} \|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2$ .  $\square$

## 4.4 Posterior predictive distribution

### Plain-language idea

To predict at a new point we don't pick one line — we average predictions over the whole posterior cloud and then add back the measurement noise. The result is again Gaussian, and its variance neatly separates “noise we can never remove” from “uncertainty because we have limited data.”

**Result/Derivation 4.5** (Predictive distribution). For a new input  $x_\star$  with  $y_\star = \langle x_\star, \theta \rangle + \varepsilon_\star$ ,  $\varepsilon_\star \sim \mathcal{N}(0, \sigma^2)$  independent of  $\theta$ , and posterior  $\theta \mid X, y \sim \mathcal{N}(\mu_N, \Sigma_N)$ ,

$$y_\star \mid x_\star, X, y \sim \mathcal{N}(\langle x_\star, \mu_N \rangle, \sigma^2 + \langle x_\star, \Sigma_N x_\star \rangle).$$

*Proof.* By the affine-transformation property of the Gaussian,  $\langle x_\star, \theta \rangle \mid X, y \sim \mathcal{N}(\langle x_\star, \mu_N \rangle, \langle x_\star, \Sigma_N x_\star \rangle)$ . Adding the independent Gaussian  $\varepsilon_\star \sim \mathcal{N}(0, \sigma^2)$  and using that the sum of independent Gaussians is Gaussian (variances add) gives the stated mean and variance. The two variance terms are the *aleatoric* noise  $\sigma^2$  and the *epistemic* term  $\langle x_\star, \Sigma_N x_\star \rangle$ , which shrinks to 0 as data accumulates.  $\square$

## 5 Logistic Regression

### 5.1 Cross-entropy loss from maximum likelihood

#### Plain-language idea

For classification we model the probability of each class rather than a number. Asking “which parameters make the observed labels most likely?” turns out to be the same as minimising the *cross-entropy* (log) loss.

**Result/Derivation 5.1** (Binary cross-entropy). With  $y \in \{-1, 1\}$ ,  $p(y = 1 \mid x; \theta) = f(x; \theta)$ , the negative log-likelihood of one example is the binary cross-entropy loss

$$\ell(f(x; \theta), y) = -[\mathbf{1}(y = 1) \ln f(x; \theta) + \mathbf{1}(y = -1) \ln(1 - f(x; \theta))].$$

*Proof.* Independence gives  $p(y \mid X; \theta) = \prod_i p(y_i \mid x_i; \theta)$  with  $p(y_i \mid x_i; \theta) = f(x_i; \theta)$  if  $y_i = 1$  and  $1 - f(x_i; \theta)$  if  $y_i = -1$ . Minimising the negative log-likelihood  $-\sum_i \ln p(y_i \mid x_i; \theta)$  is, per example, exactly the indicator sum above.  $\square$

### 5.2 The compact logistic loss

#### Plain-language idea

Plugging the sigmoid into cross-entropy and using  $\pm 1$  labels collapses the two-case formula into one clean expression  $\ln(1 + e^{-y\langle \theta, x \rangle})$ .

**Result/Derivation 5.2** (Logistic loss). For  $h(z) = \frac{1}{1+e^{-z}}$  and  $y \in \{-1, 1\}$ ,

$$\ell(h(\langle \theta, x \rangle), y) = \ln(1 + e^{-y\langle \theta, x \rangle}), \quad L(\theta) = \frac{1}{n} \sum_{i=1}^n \ln(1 + e^{-y_i \langle \theta, x_i \rangle}).$$

*Proof.* With  $p(y = 1 \mid x) = \frac{e^{\langle \theta, x \rangle}}{1 + e^{\langle \theta, x \rangle}}$  and  $p(y = -1 \mid x) = \frac{e^{-\langle \theta, x \rangle}}{1 + e^{-\langle \theta, x \rangle}}$ , both cases combine into  $p(y \mid x) = \frac{e^{y\langle \theta, x \rangle}}{1 + e^{y\langle \theta, x \rangle}}$ . Hence

$$\ell = -\ln \frac{e^{y\langle \theta, x \rangle}}{1 + e^{y\langle \theta, x \rangle}} = \ln(1 + e^{-y\langle \theta, x \rangle}). \quad \square$$

### 5.3 From binary to multiclass: the softmax

#### Plain-language idea

Binary logistic regression already hides a symmetric two-vector form; writing the probability of class 1 as a ratio of exponentials generalises in one step to any number of classes — the softmax.

**Result/Derivation 5.3** (Softmax generalisation). Binary logistic regression with  $\theta = \theta_1 - \theta_2$  can be written  $p(y = 1 | x) = \frac{e^{\langle \theta_1, x \rangle}}{e^{\langle \theta_1, x \rangle} + e^{\langle \theta_2, x \rangle}}$ , which generalises to  $p(y = c | x; \theta) = \frac{e^{\langle \theta_c, x \rangle}}{\sum_{c'=1}^C e^{\langle \theta_{c'}, x \rangle}}$ .

*Proof.* Dividing numerator and denominator of  $\frac{e^{\langle \theta_1, x \rangle}}{1 + e^{\langle \theta, x \rangle}}$  by appropriate factors and substituting  $\theta = \theta_1 - \theta_2$  yields the two-vector symmetric form. Replacing the two-term denominator by a sum over  $C$  classes gives the softmax, which maps logits in  $\mathbb{R}^C$  to probabilities in  $(0, 1)^C$  that sum to one.  $\square$

### 5.4 Cross-entropy decomposes into entropy plus KL

#### Plain-language idea

Cross-entropy between the true label distribution and the model splits into two pieces: an unavoidable part (the entropy of the truth) and a part you can actually drive down (the KL gap between model and truth). Training only affects the second.

**Result/Derivation 5.4** (CE = entropy + KL). For distributions  $q, p$  over classes,

$$H(q, p) = - \sum_c q(c) \ln p(c) = - \underbrace{\sum_c q(c) \ln q(c)}_{H(q)} + \underbrace{\sum_c q(c) \ln \frac{q(c)}{p(c)}}_{\text{KL}(q||p)},$$

with  $\text{KL}(q||p) \geq 0$  and equality iff  $q = p$ . Since  $H(q)$  is independent of the model, minimising cross-entropy is equivalent to minimising  $\text{KL}(q||p)$ .

*Proof.* Add and subtract  $\sum_c q(c) \ln q(c)$ :  $-\sum_c q(c) \ln p(c) = -\sum_c q(c) \ln q(c) + \sum_c q(c) \ln \frac{q(c)}{p(c)}$ . The first term is  $H(q)$  and the second is  $\text{KL}(q||p)$ ; non-negativity of KL is Gibbs' inequality.  $\square$

### 5.5 Logistic regression has a linear decision boundary

#### Plain-language idea

Even though logistic regression bends probabilities through a sigmoid, the cut-off between predicting one class and the other is a flat hyperplane: it is a linear classifier wearing a nonlinear probability coat.

**Result/Derivation 5.5** (Linear boundary). The rule “predict 1 iff  $p(y = 1 | x; \theta) \geq \frac{1}{2}$ ” is equivalent to “predict 1 iff  $\langle \theta, x \rangle \geq 0$ ”; the decision boundary is the hyperplane  $\langle \theta, x \rangle = 0$ .

*Proof.*  $\frac{1}{1+e^{-z}} \geq \frac{1}{2} \iff e^{-z} \leq 1 \iff z \geq 0$ . With  $z = \langle \theta, x \rangle$ , the threshold on probability becomes a threshold on the linear score, whose boundary  $\langle \theta, x \rangle = 0$  is a hyperplane.  $\square$

## 5.6 Poisson regression negative log-likelihood

### Plain-language idea

For counts we use a Poisson model with rate  $e^{\langle \theta, x \rangle}$  (the exponential keeps the rate positive). Its negative log-likelihood simplifies to a clean convex objective — a member of the generalised-linear-model family.

**Result/Derivation 5.6** (Poisson NLL). With  $y_i \mid x_i \sim \text{Pois}(\lambda_i)$ ,  $\lambda_i = \exp(\langle \theta, x_i \rangle)$ , maximum likelihood solves

$$\min_{\theta} \sum_{i=1}^n (\exp(\langle \theta, x_i \rangle) - y_i \langle \theta, x_i \rangle).$$

*Proof.* The likelihood is  $\prod_i \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!}$ . The negative log-likelihood is  $\sum_i (\lambda_i - y_i \ln \lambda_i + \ln y_i!)$ . Substituting  $\ln \lambda_i = \langle \theta, x_i \rangle$  and  $\lambda_i = e^{\langle \theta, x_i \rangle}$ , and dropping the constant  $\ln y_i!$ , leaves  $\sum_i (\exp(\langle \theta, x_i \rangle) - y_i \langle \theta, x_i \rangle)$ . The associated link is  $\phi(\mu) = \log \mu$ , since  $\mu(x) = \mathbb{E}[y \mid x] = \lambda(x) = e^{\langle \theta, x \rangle}$ .  $\square$

## 6 Loss Functions: Decision-Theoretic Foundations

### 6.1 Pointwise characterisation of the Bayes-optimal predictor

#### Plain-language idea

With infinite data, training minimises the *true* expected loss. Because the predictor's value at one input never affects its value at another, we can optimise input-by-input: at each  $x$ , pick the number that minimises the average loss against the conditional distribution of  $y$  there.

**Result/Derivation 6.1** (Pointwise minimisation). The Bayes-optimal predictor  $f^* \in \arg \min_f \mathbb{E}_{p^*(x,y)}[\ell(y, f(x))]$  satisfies, at each  $x$ ,

$$f^*(x) \in \arg \min_{t \in \mathbb{R}} \mathbb{E}_{p^*(y|x)}[\ell(y, t)].$$

*Proof.* By the tower rule,  $\mathbb{E}_{p^*(x,y)}[\ell(y, f(x))] = \mathbb{E}_{p^*(x)}[\mathbb{E}_{p^*(y|x)}[\ell(y, f(x))]]$ . The inner expectation depends on  $f$  only through the scalar  $f(x)$ , and  $f$  is unrestricted, so minimising the inner expectation separately at each  $x$  minimises the outer expectation. Hence the minimiser of the inner problem,  $f^*(x) = \arg \min_t \mathbb{E}_{p^*(y|x)}[\ell(y, t)]$ , is Bayes-optimal.  $\square$

### 6.2 Squared loss targets the conditional mean

#### Plain-language idea

If your loss is squared error, the best single guess for a noisy quantity is its average. So a squared-loss learner is implicitly estimating the conditional mean of  $y$ .

**Result/Derivation 6.2.** For  $\ell(y, t) = (y - t)^2$ , the Bayes-optimal predictor is  $f^*(x) = \mathbb{E}[y \mid x]$ .

*Proof.* The conditional risk  $\mathbb{E}[(y - t)^2 \mid x] = \mathbb{E}[y^2 \mid x] - 2t\mathbb{E}[y \mid x] + t^2$  is a strictly convex quadratic in  $t$ . Differentiating,  $\frac{\partial}{\partial t} \mathbb{E}[(y - t)^2 \mid x] = -2\mathbb{E}[y \mid x] + 2t = 0$  gives the unique minimiser  $t = \mathbb{E}[y \mid x]$ .  $\square$

### 6.3 Absolute loss targets the conditional median

#### Plain-language idea

Swap squared error for absolute error and the optimal guess changes from the mean to the median — the value with equal probability mass above and below. This is why absolute loss is more robust to outliers. The derivation needs a little care because  $|y - t|$  has a kink, so we differentiate the integral using the Leibniz rule.

**Result/Derivation 6.3.** For  $\ell(y, t) = |y - t|$ , any Bayes-optimal predictor is a conditional median:  $\mathbb{P}(y \leq f^*(x) \mid x) = \frac{1}{2}$ .

*Proof.* Split the conditional risk at  $t$ :

$$\mathbb{E}[|y - t| \mid x] = \int_{-\infty}^t (t - y)p(y \mid x) dy + \int_t^{\infty} (y - t)p(y \mid x) dy.$$

Differentiating each term by the Leibniz rule (the boundary terms vanish because the integrand is zero at  $y = t$ ):

$$\begin{aligned} \frac{d}{dt} \int_{-\infty}^t (t - y)p(y \mid x) dy &= \int_{-\infty}^t p(y \mid x) dy = \mathbb{P}(y \leq t \mid x), \\ \frac{d}{dt} \int_t^{\infty} (y - t)p(y \mid x) dy &= - \int_t^{\infty} p(y \mid x) dy = -\mathbb{P}(y > t \mid x). \end{aligned}$$

Adding,  $\frac{d}{dt} \mathbb{E}[|y - t| \mid x] = \mathbb{P}(y \leq t \mid x) - \mathbb{P}(y > t \mid x) = 2\mathbb{P}(y \leq t \mid x) - 1$ . Setting this to zero gives  $\mathbb{P}(y \leq t^* \mid x) = \frac{1}{2}$ , the median.  $\square$

### 6.4 Negative log-likelihood is a proper loss

#### Plain-language idea

A loss is “proper” if, given infinite data, the best predictor recovers the *true* probability distribution rather than just a summary. NLL is proper because its population risk is the true entropy (fixed) plus a KL gap that is zero exactly when the model matches the truth.

**Result/Derivation 6.4** (NLL properness). For  $\ell(y, f(x)) = -\log p_f(y \mid x)$ , the population risk is

$$\mathbb{E}_{p^*(y|x)}[-\log p_f(y \mid x)] = \underbrace{H(p^*)}_{\text{fixed}} + \underbrace{\text{KL}(p^*(\cdot \mid x) \parallel p_f(\cdot \mid x))}_{\geq 0},$$

minimised iff  $p_f(\cdot \mid x) = p^*(\cdot \mid x)$ . Hence NLL is proper (though not necessarily *strictly* proper).

*Proof.* For fixed  $x$ , add and subtract  $\log p^*(y \mid x)$  inside the expectation:

$$\mathbb{E}_{p^*(y|x)}[-\log p_f(y \mid x)] = \mathbb{E}_{p^*(y|x)}[-\log p^*(y \mid x)] + \mathbb{E}_{p^*(y|x)}\left[\log \frac{p^*(y|x)}{p_f(y|x)}\right].$$

The first term is the entropy  $H(p^*)$ , independent of  $f$ ; the second is  $\text{KL}(p^* \parallel p_f) \geq 0$ , zero iff the densities agree a.e. So the minimiser recovers  $p^*(\cdot \mid x)$ .  $\square$

### 6.5 0–1 loss: Bayes classifier and impropriety

### Plain-language idea

Under 0–1 loss the optimal rule just predicts the majority class — it recovers the *boundary* ( $\eta$  above or below  $\frac{1}{2}$ ) but throws away the actual probability  $\eta(x)$ . So 0–1 loss is *not* proper: it cannot tell a confident 90% from a hesitant 51%.

**Result/Derivation 6.5.** For  $y \in \{-1, +1\}$  and  $\ell(y, f(x)) = \mathbf{1}(y \neq f(x))$ ,  $f^*(x) = \text{sign}(\eta(x) - \frac{1}{2})$ , with minimal conditional error  $\min\{\eta(x), 1 - \eta(x)\}$ . The map is non-invertible in  $\eta$ , so 0–1 loss is not proper.

*Proof.*  $\mathbb{E}_{p^*(y|x)}[\mathbf{1}(y \neq t)] = \mathbb{P}(y \neq t \mid x)$ , minimised over  $t \in \{-1, +1\}$  by choosing the more probable label, i.e.  $t = +1$  when  $\eta(x) > \frac{1}{2}$  and  $t = -1$  when  $\eta(x) < \frac{1}{2}$ , giving  $f^*(x) = \text{sign}(\eta(x) - \frac{1}{2})$ . Since  $f^*$  depends on  $\eta$  only through its sign relative to  $\frac{1}{2}$ , distinct  $\eta$  values map to the same prediction; the true  $\eta(x)$  cannot be recovered.  $\square$

## 6.6 Cross-entropy and logistic losses are strictly proper

### Plain-language idea

Unlike 0–1 loss, cross-entropy is minimised *only* by reporting the true probability  $\eta(x)$  itself, and that minimiser is unique — it is strictly proper. The logistic (margin) loss is the same story viewed through the log-odds.

**Result/Derivation 6.6** (Cross-entropy strict properness). For  $\ell(y, t)$  with  $t \in (0, 1)$ , the cross-entropy conditional risk  $\eta(x)(-\log t) + (1 - \eta(x))(-\log(1 - t))$  is uniquely minimised at  $t^* = \eta(x) = p^*(y = 1 \mid x)$ .

*Proof.* Differentiate in  $t$ :  $-\frac{\eta(x)}{t} + \frac{1-\eta(x)}{1-t} = 0 \Rightarrow \eta(x)(1-t) = (1-\eta(x))t \Rightarrow t = \eta(x)$ . The objective is strictly convex on  $(0, 1)$ , so this minimiser is unique; the optimal predictor recovers the full conditional distribution, hence cross-entropy is strictly proper.  $\square$

**Result/Derivation 6.7** (Logistic loss strict properness). For the margin form  $\ell(y, f(x)) = \log(1 + e^{-yf(x)})$ ,  $f^*(x) = \log \frac{\eta(x)}{1-\eta(x)}$ .

*Proof.* Fix  $x$  and write  $\varphi(t) = \eta(x) \log(1 + e^{-t}) + (1 - \eta(x)) \log(1 + e^t)$ . Then

$$\varphi'(t) = -\frac{\eta(x)}{1 + e^t} + \frac{(1 - \eta(x))e^t}{1 + e^t}.$$

Setting  $\varphi'(t) = 0$  gives  $\eta(x) = (1 - \eta(x))e^t$ , i.e.  $e^t = \frac{\eta(x)}{1-\eta(x)}$ , so  $f^*(x) = \log \frac{\eta(x)}{1-\eta(x)}$ . This log-odds map is invertible in  $\eta$ , hence logistic loss is strictly proper. (By contrast, hinge loss gives  $f^*(x) = \text{sign}(\eta(x) - \frac{1}{2})$ , which is not invertible and hence not strictly proper.)  $\square$

## 7 Loss Functions: Regression and Classification

### 7.1 Noise model determines the regression loss

#### Plain-language idea

In regression, the loss you should use is dictated by what you assume about the noise: light-tailed Gaussian noise gives squared loss; heavy-tailed Laplace noise gives absolute loss (which shrugs off outliers). The bridge is always the same: loss = negative log-density of the noise.

**Result/Derivation 7.1** (Gaussian  $\Rightarrow$  squared, Laplace  $\Rightarrow$  absolute). With  $y = f(x) + \varepsilon$  and  $p_f(y | x) = p_\varepsilon(y - f(x))$ , the NLL loss  $\ell(y, f(x)) = -\log p_\varepsilon(y - f(x))$  gives:  $\varepsilon \sim \mathcal{N}(0, \sigma^2) \Rightarrow \ell \propto (y - f(x))^2$ ;  $\varepsilon \sim \text{Laplace}(0, b) \Rightarrow \ell \propto |y - f(x)|$ .

*Proof.* Gaussian:  $-\log p_\varepsilon(t) = \frac{t^2}{2\sigma^2} + \text{const}$ , so with  $t = y - f(x)$  the loss is proportional to  $(y - f(x))^2$ . Laplace:  $p_\varepsilon(t) = \frac{1}{2b}e^{-|t|/b}$  gives  $-\log p_\varepsilon(t) = \frac{|t|}{b} + \text{const}$ , proportional to  $|y - f(x)|$ . The quadratic grows fast (outlier-sensitive); the absolute grows linearly (robust).  $\square$

## 7.2 Distance to the decision boundary and the margin

### Plain-language idea

For a linear classifier the score  $\langle \theta, x \rangle$  is, up to the length of  $\theta$ , just the signed distance from the point to the decision hyperplane. Multiplying by the label gives the *margin*: positive means correct, larger means more confident.

**Result/Derivation 7.2** (Distance formula). For  $g(x; \theta) = \langle \theta, x \rangle$  with boundary  $\langle \theta, x \rangle = 0$ ,

$$\text{dist}(x, \text{boundary}) = \frac{|\langle \theta, x \rangle|}{\|\theta\|_2}, \quad \text{so} \quad |g(x; \theta)| = \|\theta\|_2 \cdot \text{dist}(x, \text{boundary}).$$

*Proof.*  $\theta$  is normal to the hyperplane. Take any boundary point  $x_0$  with  $\langle \theta, x_0 \rangle = 0$ . Project the displacement  $x - x_0$  onto the unit normal  $\theta/\|\theta\|_2$ :  $\text{dist} = \frac{|\langle \theta, x - x_0 \rangle|}{\|\theta\|_2} = \frac{|\langle \theta, x \rangle|}{\|\theta\|_2}$ , using  $\langle \theta, x_0 \rangle = 0$ . Rearranging gives the second equality. The *margin*  $m = yg(x)$  then has sign indicating correctness and magnitude proportional to distance.  $\square$

## 8 Evaluating Performance

### 8.1 Hold-out validation is unbiased

#### Plain-language idea

If you evaluate a model on data it never trained on, your error estimate is, on average, exactly the true generalisation error. The catch (variance vs training-size) is practical, not a bias.

**Result/Derivation 8.1** (Unbiasedness). If the validation set  $V$  is independent of the training set  $D$  and drawn from the same  $p_{x,y}$ , then  $\mathbb{E}[E_{\text{hold-out}}] = E^*$ , where  $E_{\text{hold-out}} = \frac{1}{m} \sum_{j=1}^m e(f(\tilde{x}_j; D), \tilde{y}_j)$ .

*Proof.* Condition on  $D$  (so  $f(\cdot; D)$  is fixed). Each validation pair  $(\tilde{x}_j, \tilde{y}_j) \sim p_{x,y}$  independently of  $D$ , so  $\mathbb{E}[e(f(\tilde{x}_j; D), \tilde{y}_j) | D] = \mathbb{E}_{(x,y) \sim p_{x,y}}[e(f(x; D), y)] = E^*(D)$ . Averaging over the  $m$  i.i.d. validation points keeps this value, and taking the expectation over  $D$  gives  $\mathbb{E}[E_{\text{hold-out}}] = \mathbb{E}_D[E^*(D)] = E^*$ .  $\square$

### 8.2 ROC-AUC equals a pairwise ranking probability

#### Plain-language idea

The area under the ROC curve has a beautifully simple meaning: pick one random positive and one random negative example; AUC is the probability the classifier scores the positive higher. A perfect ranker scores 1; a coin flip scores 0.5.



**Result/Derivation 8.2** (AUC as ranking probability). With positive scores  $\{s_i^+\}_{i=1}^{n_+}$  and negative scores  $\{s_j^-\}_{j=1}^{n_-}$ ,

$$\text{AUC} = \frac{1}{n_+ n_-} \sum_{i=1}^{n_+} \sum_{j=1}^{n_-} \left[ \mathbf{1}(s_i^+ > s_j^-) + \frac{1}{2} \mathbf{1}(s_i^+ = s_j^-) \right] = \frac{\#\{\text{correctly ordered pairs}\}}{n_+ n_-}.$$

*Proof (sketch, as in the lecture).* Sweep the threshold  $r$  from high to low; points enter the prediction in decreasing score order. Each newly admitted positive raises the true-positive rate (a vertical step), each negative raises the false-positive rate (a horizontal step). The ROC curve is this staircase, and its area accumulates, for each positive, the fraction of negatives ranked below it: a positive at rank above a negative contributes one unit of area (ties contribute one half). Summing over all  $n_+ n_-$  pairs and normalising gives the stated formula. A perfect classifier has every positive scored above every negative (AUC = 1); random scoring orders each pair correctly with probability  $\frac{1}{2}$  (AUC = 0.5).  $\square$

## 9 Model Complexity and Generalisation: Classical View

### 9.1 The bias–variance decomposition of an estimator

#### Plain-language idea

Any estimator’s average squared error breaks cleanly into two parts: how far off it is on average (bias) and how much it jiggles around its own average (variance). The cross term conveniently cancels.

**Result/Derivation 9.1** (General decomposition). For a random estimator  $z$  of a constant  $z_0$ ,

$$\mathbb{E}[(z - z_0)^2] = \underbrace{\mathbb{E}[(z - \mathbb{E}[z])^2]}_{\text{Variance}} + \underbrace{(\mathbb{E}[z] - z_0)^2}_{\text{Bias}^2}.$$

*Proof.* Write  $z - z_0 = (z - \mathbb{E}[z]) + (\mathbb{E}[z] - z_0)$  and square:

$$(z - z_0)^2 = (z - \mathbb{E}[z])^2 + 2(z - \mathbb{E}[z])(\mathbb{E}[z] - z_0) + (\mathbb{E}[z] - z_0)^2.$$

Taking expectations, the middle term has factor  $\mathbb{E}[z - \mathbb{E}[z]] = 0$ , leaving  $\mathbb{E}[(z - \mathbb{E}[z])^2] + (\mathbb{E}[z] - z_0)^2$ .  $\square$

### 9.2 Bias–variance decomposition of the generalisation error

#### Plain-language idea

Apply the same split to a learned regressor, treating the training set itself as random. The expected test error becomes squared bias (how well the *average* learned model matches the truth) + variance (sensitivity to which dataset you happened to draw) + irreducible noise (the part no model can ever remove).

#### Approach

Insert the data model  $y = f^*(x) + \varepsilon$ , condition on  $x$  to peel off the noise term using  $\mathbb{E}[\varepsilon | x] = 0$ , then apply the general decomposition with  $z = f(x; D)$ ,  $z_0 = f^*(x)$  and the average model  $\bar{f}(x) = \mathbb{E}_D[f(x; D)]$ .

**Theorem 9.2** (Bias–variance for regression). With  $y = f^*(x) + \varepsilon$ ,  $\mathbb{E}[\varepsilon | x] = 0$ ,  $\text{Var}(\varepsilon | x) = \sigma^2$ , and  $\bar{f}(x) = \mathbb{E}_D[f(x; D)]$ ,

$$\bar{E}^* = \underbrace{\mathbb{E}_x[(\bar{f}(x) - f^*(x))^2]}_{\text{Bias}^2} + \underbrace{\mathbb{E}_x[\mathbb{E}_D[(f(x; D) - \bar{f}(x))^2]]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible}}.$$

*Proof.* Start from  $\bar{E}^* = \mathbb{E}_D[\mathbb{E}_{(x, \varepsilon)}[(f(x; D) - f^*(x) - \varepsilon)^2]]$ . Condition on  $x$  and expand the inner expectation over  $\varepsilon$ :

$$\mathbb{E}_{\varepsilon|x}[(f - f^* - \varepsilon)^2] = (f - f^*)^2 - 2(f - f^*)\mathbb{E}[\varepsilon | x] + \mathbb{E}[\varepsilon^2 | x] = (f - f^*)^2 + \sigma^2,$$

using  $\mathbb{E}[\varepsilon | x] = 0$ . Hence  $\bar{E}^* = \mathbb{E}_D\mathbb{E}_x[(f(x; D) - f^*(x))^2] + \sigma^2$ . For fixed  $x$ , apply the general decomposition with  $z = f(x; D)$ ,  $z_0 = f^*(x)$ :

$$\mathbb{E}_D[(f(x; D) - f^*(x))^2] = (\bar{f}(x) - f^*(x))^2 + \mathbb{E}_D[(f(x; D) - \bar{f}(x))^2].$$

Averaging over  $x$  and adding  $\sigma^2$  gives the result.  $\square$

### 9.3 Heuristic links: training error and the generalisation gap

#### Plain-language idea

You can't measure bias and variance directly (they need the unknown true distribution), but you can measure training error and the train–test gap. These line up: large training error signals high bias; a large gap signals high variance.

**Result/Derivation 9.3** (Bias bounded by training error).  $E_{\text{train}} \lesssim \text{Bias}^2 + \sigma^2$ , and consequently the generalisation gap  $\bar{E}^* - E_{\text{train}} \gtrsim \text{Variance}$ .

*Proof.* For the average model,  $\mathbb{E}_x[(\bar{f}(x) - y)^2] = \sigma^2 + \mathbb{E}_x[(\bar{f}(x) - f^*(x))^2] = \sigma^2 + \text{Bias}^2$  (the cross term vanishes since  $\mathbb{E}[\varepsilon | x] = 0$ ). Approximating the expectation by the training average and using that training minimises squared error,  $\frac{1}{n} \sum_i (\bar{f}(x_i) - y_i)^2 \geq \frac{1}{n} \sum_i (f(x_i; D) - y_i)^2 = E_{\text{train}}$ , so  $E_{\text{train}} \lesssim \sigma^2 + \text{Bias}^2$ . Subtracting from  $\bar{E}^* = \sigma^2 + \text{Bias}^2 + \text{Variance}$  leaves  $\bar{E}^* - E_{\text{train}} \gtrsim \text{Variance}$ .  $\square$

*Modern view (no full proof).* The course's “modern” lecture exhibits *double descent*: for highly overparameterised models the test error can fall again past the interpolation threshold. It is illustrated with the closed-form risk of minimum-norm least squares and the observation that bias appears only when  $p > n$  (the part of  $\theta^*$  outside the row space of  $X$  cannot be recovered), but no theorem is proved.

## 10 Optimisation

### 10.1 Adam's bias correction

#### Plain-language idea

Adam tracks a running (exponential) average of gradients, started at zero. At the very beginning that average is biased toward zero — it hasn't “warmed up.” Dividing by the missing weight  $1 - \beta_1^k$  removes exactly this start-up bias, and likewise for the squared-gradient average.

**Result/Derivation 10.1** (Bias correction). With  $m_0 = 0$  and  $m_k = \beta_1 m_{k-1} + (1 - \beta_1)g_k$ , if the gradients have roughly constant mean  $\mathbb{E}[g_s] \approx \mu$ , then  $\mathbb{E}[m_k] \approx \mu(1 - \beta_1^k)$ , so the unbiased estimate is  $\hat{m}_k = \frac{m_k}{1 - \beta_1^k}$  (and likewise  $\hat{v}_k = \frac{v_k}{1 - \beta_2^k}$ ).

*Proof.* Unrolling the recursion with  $m_0 = 0$ ,

$$m_k = (1 - \beta_1) \sum_{s=1}^k \beta_1^{k-s} g_s.$$

Taking expectations with  $\mathbb{E}[g_s] \approx \mu$  and summing the geometric series,

$$\mathbb{E}[m_k] \approx (1 - \beta_1) \mu \sum_{s=1}^k \beta_1^{k-s} = (1 - \beta_1) \mu \cdot \frac{1 - \beta_1^k}{1 - \beta_1} = \mu(1 - \beta_1^k).$$

Since  $1 - \beta_1^k < 1$ ,  $m_k$  underestimates  $\mu$  early on; dividing by  $1 - \beta_1^k$  corrects it. The identical argument applies to the second-moment estimate  $v_k$ .  $\square$

## 11 Regularisation

### 11.1 Bias and variance of the ridge estimator

#### Plain-language idea

Ridge deliberately shrinks coefficients toward zero. That shrinkage introduces bias (the expected estimate is no longer the truth) but suppresses the noise term, lowering variance. Increasing the penalty  $\lambda$  trades more of one for less of the other.

**Result/Derivation 11.1** (Ridge bias). Under  $y = X\theta^* + \varepsilon$  with  $\mathbb{E}[\varepsilon] = 0$ , the ridge estimator  $\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$  has  $\mathbb{E}[\hat{\theta}_\lambda] = S_\lambda \theta^*$  and  $\text{Bias}(\hat{\theta}_\lambda) = (S_\lambda - I)\theta^*$ , where  $S_\lambda = (X^\top X + \lambda I)^{-1} X^\top X$ .

*Proof.* Substituting the model,  $\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top X \theta^* + (X^\top X + \lambda I)^{-1} X^\top \varepsilon$ . Taking expectations kills the second term, leaving  $\mathbb{E}[\hat{\theta}_\lambda] = S_\lambda \theta^*$ . The bias is  $\mathbb{E}[\hat{\theta}_\lambda] - \theta^* = (S_\lambda - I)\theta^*$ , nonzero for  $\lambda > 0$  and growing with  $\lambda$ .  $\square$

**Result/Derivation 11.2** (Ridge variance). With  $\text{Cov}(\varepsilon) = \sigma^2 I$ ,

$$\text{Cov}(\hat{\theta}_\lambda) = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1},$$

which shrinks as  $\lambda$  grows.

*Proof.* The fluctuation is  $\hat{\theta}_\lambda - \mathbb{E}[\hat{\theta}_\lambda] = (X^\top X + \lambda I)^{-1} X^\top \varepsilon$ . Hence

$$\text{Cov}(\hat{\theta}_\lambda) = (X^\top X + \lambda I)^{-1} X^\top \mathbb{E}[\varepsilon \varepsilon^\top] X (X^\top X + \lambda I)^{-1} = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1},$$

using  $\mathbb{E}[\varepsilon \varepsilon^\top] = \sigma^2 I$ . Larger  $\lambda$  makes  $(X^\top X + \lambda I)^{-1}$  smaller, damping the variance. The prediction error at  $x_0$  then decomposes as  $\mathbb{E}[(\hat{y}_0 - y_0)^2] = (x_0^\top \text{Bias}(\hat{\theta}_\lambda))^2 + x_0^\top \text{Cov}(\hat{\theta}_\lambda) x_0 + \sigma^2$ .  $\square$

### 11.2 Variance of a linear combination

#### Plain-language idea

A small but constantly-used fact: the variance of a fixed linear readout of a random vector is a quadratic form in that vector's covariance.

**Lemma 11.3.** For fixed  $x$  and random  $\theta$  with  $\mu = \mathbb{E}[\theta]$ ,  $\Sigma = \text{Cov}(\theta)$ , the scalar  $y = x^\top \theta$  has  $\text{Var}(y) = x^\top \Sigma x$ .

*Proof.*  $\text{Var}(x^\top \theta) = \mathbb{E}[(x^\top \theta - x^\top \mu)^2] = \mathbb{E}[(x^\top (\theta - \mu))^2] = \mathbb{E}[x^\top (\theta - \mu)(\theta - \mu)^\top x] = x^\top \mathbb{E}[(\theta - \mu)(\theta - \mu)^\top] x = x^\top \Sigma x$ .  $\square$

## 12 Kernel Methods: Kernel Ridge Regression

### 12.1 Kernelising ridge regression via the push-through identity

#### Plain-language idea

Ridge regression in a huge (even infinite) feature space looks hopeless — you'd have to invert a giant feature-by-feature matrix. A matrix identity lets us flip the computation so it only ever needs an  $n \times n$  matrix of *inner products* between data points. Replacing those inner products by a kernel means we never build the features at all.

**Result/Derivation 12.1** (Kernelised prediction). For ridge in feature space, the prediction  $\hat{y}(x_*) = \phi(x_*)^\top (\Phi^\top \Phi + n\lambda I_p)^{-1} \Phi^\top y$  can be rewritten as

$$\hat{y}(x_*) = k_*^\top (K + n\lambda I_n)^{-1} y, \quad K = \Phi \Phi^\top, \quad [k_*]_i = k(x_i, x_*) = \phi(x_i)^\top \phi(x_*).$$

*Proof.* The push-through identity  $(A^\top A + \lambda I)^{-1} A^\top = A^\top (AA^\top + \lambda I)^{-1}$ , applied with  $A = \Phi$ , gives  $(\Phi^\top \Phi + n\lambda I_p)^{-1} \Phi^\top = \Phi^\top (\Phi \Phi^\top + n\lambda I_n)^{-1}$ . Therefore

$$\hat{y}(x_*) = \phi(x_*)^\top \Phi^\top (\Phi \Phi^\top + n\lambda I_n)^{-1} y = k_*^\top (K + n\lambda I_n)^{-1} y,$$

since  $\phi(x_*)^\top \Phi^\top = k_*^\top$  and  $\Phi \Phi^\top = K$ . Only an  $n \times n$  inverse and kernel evaluations are needed;  $\phi$  is never formed.  $\square$

### 12.2 When is a kernel a valid inner product? PSD examples

#### Plain-language idea

A function  $k$  can stand in for an inner product  $\phi(x)^\top \phi(x')$  exactly when it is positive semidefinite (every Gram matrix it builds is PSD). The slides prove PSD directly for two concrete kernels by writing the quadratic form as a sum of squares.

**Theorem 12.2** (Moore–Aronszajn — stated). A symmetric kernel  $k$  is PSD if and only if there exists a (possibly infinite-dimensional) feature map  $\phi$  with  $k(x, x') = \phi(x)^\top \phi(x')$ . The associated function space is the Reproducing Kernel Hilbert Space (RKHS).

**Result/Derivation 12.3** (Polynomial kernel is PSD).  $k(x, x') = (1 + xx')^2$  is PSD, with feature map  $\phi(x) = (1, \sqrt{2}x, x^2)^\top$ .

*Proof.* Expand  $(1 + xx')^2 = 1 + 2xx' + x^2x'^2 = \phi(x)^\top \phi(x')$ . For any  $v \in \mathbb{R}^n$ ,

$$v^\top K_n v = \sum_{i,j} v_i v_j (1 + 2x_i x_j + x_i^2 x_j^2) = \left( \sum_i v_i \right)^2 + 2 \left( \sum_i v_i x_i \right)^2 + \left( \sum_i v_i x_i^2 \right)^2 \geq 0.$$

$\square$

**Result/Derivation 12.4** (Exponential kernel is PSD).  $k(x, x') = e^{xx'}$  is PSD, with infinite feature map  $\phi(x) = (1, x, \frac{x^2}{\sqrt{2!}}, \frac{x^3}{\sqrt{3!}}, \dots)$ .

*Proof.* By the Taylor series  $e^{xx'} = \sum_{m=0}^{\infty} \frac{(xx')^m}{m!} = \sum_{m=0}^{\infty} \left( \frac{x^m}{\sqrt{m!}} \right) \left( \frac{x'^m}{\sqrt{m!}} \right)$ . Then

$$v^\top K_n v = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{i,j} v_i v_j x_i^m x_j^m = \sum_{m=0}^{\infty} \frac{1}{m!} \left( \sum_i v_i x_i^m \right)^2 \geq 0,$$

a non-negative weighted sum of squares.  $\square$

### 12.3 The Representer Theorem (finite-dimensional case)

### Plain-language idea

Even if the feature space is enormous, the best solution always lives in the tiny subspace spanned by the training points. Any component pointing “away” from the data does nothing for the fit but costs extra in the  $\ell_2$  penalty, so the optimiser sets it to zero. This is what makes kernel methods finite-dimensional in practice.

**Theorem 12.5** (Representer Theorem, finite-dimensional). For any loss  $\ell$ , any minimiser  $\hat{\theta}$  of

$$\min_{\theta \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \ell(y_i, \phi(x_i)^\top \theta) + \lambda \|\theta\|_2^2$$

has the form  $\hat{\theta} = \sum_{i=1}^n \alpha_i \phi(x_i)$  for some  $\alpha \in \mathbb{R}^n$ .

*Proof.* Decompose any  $\theta = \theta_{\parallel} + \theta_{\perp}$ , where  $\theta_{\parallel} \in \text{span}\{\phi(x_1), \dots, \phi(x_n)\}$  and  $\theta_{\perp} \perp \phi(x_i)$  for all  $i$ . For each training point,

$$\phi(x_i)^\top \theta = \phi(x_i)^\top \theta_{\parallel} + \phi(x_i)^\top \theta_{\perp} = \phi(x_i)^\top \theta_{\parallel},$$

so the loss term depends only on  $\theta_{\parallel}$ . For the penalty, by orthogonality (Pythagoras),  $\|\theta\|_2^2 = \|\theta_{\parallel}\|_2^2 + \|\theta_{\perp}\|_2^2 \geq \|\theta_{\parallel}\|_2^2$ , with equality iff  $\theta_{\perp} = 0$ . Thus removing  $\theta_{\perp}$  never increases the objective, so any minimiser has  $\hat{\theta}_{\perp} = 0$ , i.e.  $\hat{\theta} \in \text{span}\{\phi(x_i)\}$ . The predictor becomes  $\hat{y}(x_{\star}) = \sum_i \alpha_i \phi(x_{\star})^\top \phi(x_i) = \sum_i \alpha_i k(x_i, x_{\star})$ .  $\square$

### Plain-language idea

The proof leans on the  $\ell_2$  penalty: Pythagoras is what guarantees discarding  $\theta_{\perp}$  helps. With an  $\ell_1$  penalty,  $\|\theta\|_1 \neq \|\theta_{\parallel}\|_1 + \|\theta_{\perp}\|_1$  in general, and the representer property can fail.

## 13 Kernel Methods: RKHS

### 13.1 The Representer Theorem (RKHS form)

#### Plain-language idea

Lifting the previous result into function space: when we optimise directly over an infinite-dimensional RKHS of functions, the optimum is still a finite sum of “kernel bumps” centred at the training points. We reason in the big space but compute with  $n$  numbers.

**Theorem 13.1** (Representer Theorem, RKHS). For any loss  $\ell$ , any minimiser  $\hat{f}$  of  $\min_{f \in \mathcal{H}_k} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)) + \lambda \|f\|_{\mathcal{H}_k}^2$  has the form  $\hat{f}(\cdot) = \sum_{i=1}^n \alpha_i k(x_i, \cdot)$ , with RKHS norm  $\|\hat{f}\|_{\mathcal{H}_k}^2 = \sum_{i,j} \alpha_i \alpha_j k(x_i, x_j) = \alpha^\top K \alpha$ .

*Proof.* The reproducing property  $f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}_k}$  plays the role of the inner product  $\phi(x)^\top \theta$  in the finite case. Decompose  $f = f_{\parallel} + f_{\perp}$  with  $f_{\parallel} \in \text{span}\{k(x_i, \cdot)\}$  and  $f_{\perp}$  orthogonal to each  $k(x_i, \cdot)$ . Then  $f(x_i) = \langle f, k(x_i, \cdot) \rangle = \langle f_{\parallel}, k(x_i, \cdot) \rangle = f_{\parallel}(x_i)$ , so the loss sees only  $f_{\parallel}$ , while  $\|f\|_{\mathcal{H}_k}^2 = \|f_{\parallel}\|_{\mathcal{H}_k}^2 + \|f_{\perp}\|_{\mathcal{H}_k}^2$ . Hence  $\hat{f}_{\perp} = 0$ . Writing  $\hat{f} = \sum_i \alpha_i k(x_i, \cdot)$  and using bilinearity of the inner product with the reproducing property gives  $\|\hat{f}\|_{\mathcal{H}_k}^2 = \sum_{i,j} \alpha_i \alpha_j \langle k(x_i, \cdot), k(x_j, \cdot) \rangle = \sum_{i,j} \alpha_i \alpha_j k(x_i, x_j) = \alpha^\top K \alpha$ .  $\square$

For kernel ridge regression specifically, substituting  $f(\cdot) = \sum_j \alpha_j k(x_j, \cdot)$  (so  $f = K\alpha$  at training points,  $\|f\|_{\mathcal{H}_k}^2 = \alpha^\top K \alpha$ ) reduces the problem to  $\min_{\alpha} \frac{1}{n} \|y - K\alpha\|_2^2 + \lambda \alpha^\top K \alpha$ , with solution  $\alpha = (K + n\lambda I)^{-1} y$  and prediction  $\hat{f}(x) = k(x)^\top (K + n\lambda I)^{-1} y$ .

## 13.2 Bias and variance of kernel ridge regression

### Plain-language idea

Writing the KRR fit at the training points as a linear “smoother”  $\hat{f} = Sy$  exposes its statistics directly: the bias is the smoother failing to be the identity, and the variance is the smoother passing the noise through (twice, since covariance is quadratic).

**Result/Derivation 13.2** (KRR bias). With  $y = f^* + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$  and the smoother  $S = K(K + n\lambda I)^{-1}$ ,  $\hat{f} = Sy$  has Bias  $= (S - I)f^* = -n\lambda(K + n\lambda I)^{-1}f^*$ .

*Proof.*  $\mathbb{E}[\hat{f}] = S\mathbb{E}[y] = Sf^*$ , so Bias  $= Sf^* - f^* = (S - I)f^*$ . For the closed form,  $S - I = K(K + n\lambda I)^{-1} - I$ , and multiplying by  $(K + n\lambda I)$  gives  $(S - I)(K + n\lambda I) = K - (K + n\lambda I) = -n\lambda I$ , hence  $S - I = -n\lambda(K + n\lambda I)^{-1}$ .  $\square$

**Result/Derivation 13.3** (KRR variance).  $\text{Cov}(\hat{f}) = \sigma^2 SS^\top = \sigma^2 S^2$  (since  $S$  is symmetric), and the training-point MSE is  $\mathbb{E}\|\hat{f} - f^*\|_2^2 = \|(S - I)f^*\|_2^2 + \sigma^2 \text{tr}(S^2)$ .

*Proof.*  $\text{Cov}(\hat{f}) = \text{Cov}(Sy) = S \text{Cov}(y) S^\top = \sigma^2 SS^\top = \sigma^2 S^2$ . Then  $\mathbb{E}\|\hat{f} - f^*\|_2^2 = \|\mathbb{E}[\hat{f}] - f^*\|_2^2 + \text{tr} \text{Cov}(\hat{f}) = \|(S - I)f^*\|_2^2 + \sigma^2 \text{tr}(S^2)$ . Larger  $\lambda$  increases the bias term and decreases the variance term. (Spectrally, with  $K = U\Lambda U^\top$ ,  $S = \sum_i \frac{\mu_i}{\mu_i + n\lambda} u_i u_i^\top$  shrinks each eigendirection by  $\frac{\mu_i}{\mu_i + n\lambda}$ .)  $\square$

## 14 Kernel Methods: Support Vector Regression & Classification

### 14.1 Weak duality

#### Plain-language idea

For any constrained minimisation, the Lagrangian dual gives a quantity that can never exceed the true optimum — a guaranteed lower bound, no convexity required. This is the foundation that lets us solve SVR/SVM through their duals.

**Theorem 14.1** (Weak duality). For  $\min_\omega f(\omega)$  s.t.  $g_i(\omega) \leq 0$  with Lagrangian  $L(\omega, \alpha) = f(\omega) + \sum_i \alpha_i g_i(\omega)$  and dual  $g(\alpha) = \min_\omega L(\omega, \alpha)$ ,

$$\max_{\alpha \geq 0} g(\alpha) \leq \min_{g_i(\omega) \leq 0} f(\omega).$$

*Proof.* Fix  $\alpha \geq 0$  and any feasible  $\omega$  (so  $g_i(\omega) \leq 0$ ). Then  $\sum_i \alpha_i g_i(\omega) \leq 0$ , hence  $L(\omega, \alpha) = f(\omega) + \sum_i \alpha_i g_i(\omega) \leq f(\omega)$ . Taking the minimum over  $\omega$  on the left,  $g(\alpha) = \min_\omega L(\omega, \alpha) \leq L(\omega, \alpha) \leq f(\omega)$  for every feasible  $\omega$ . Thus  $g(\alpha)$  lower-bounds  $f$  on the feasible set; maximising over  $\alpha \geq 0$  preserves the inequality. (When the problem is convex and Slater’s condition holds, equality — strong duality — holds; SVR/SVM have affine constraints, so this applies.)  $\square$

### 14.2 The SVR dual

#### Plain-language idea

Support vector regression replaces squared loss with an  $\varepsilon$ -tube: errors inside the tube cost nothing. Dualising the problem expresses the solution purely through kernels and reveals *sparsity* — only points on or outside the tube get nonzero weight (the support vectors).

### Approach

Introduce slack variables and multipliers, form the Lagrangian, then (i) set  $\partial_\theta L = 0$  to express  $\hat{\theta}$  in the data span, (ii) substitute back to get a kernel quadratic, (iii) set  $\partial_\xi L = 0$  to turn the slack multipliers into box constraints, and collect terms.

**Result/Derivation 14.2** (SVR dual). The SVR primal  $\min_{\theta, \xi} \frac{1}{n} \sum_i (\xi_i^+ + \xi_i^-) + \lambda \|\theta\|^2$  s.t.  $\xi_i^+ \geq y_i - \theta^\top \phi(x_i) - \varepsilon$ ,  $\xi_i^- \geq \theta^\top \phi(x_i) - y_i - \varepsilon$ ,  $\xi_i^\pm \geq 0$ , has stationary point  $\hat{\theta} = \sum_i \alpha_i \phi(x_i)$  with  $\alpha_i = \alpha_i^+ - \alpha_i^-$ , and (after rescaling) dual

$$\min_{\alpha} \frac{1}{2} \alpha^\top K \alpha - \alpha^\top y + \varepsilon \|\alpha\|_1 \quad \text{s.t.} \quad |\alpha_i| \leq \frac{1}{2\lambda n}.$$

*Proof.* With multipliers  $\alpha_i^\pm, \gamma_i^\pm \geq 0$ ,

$$L = \frac{1}{n} \sum_i (\xi_i^+ + \xi_i^-) + \lambda \|\theta\|^2 + \sum_i \alpha_i^+ (y_i - \theta^\top \phi(x_i) - \varepsilon - \xi_i^+) + \sum_i \alpha_i^- (\theta^\top \phi(x_i) - y_i - \varepsilon - \xi_i^-) - \sum_i \gamma_i^+ \xi_i^+ - \sum_i \gamma_i^- \xi_i^-.$$

*Stationarity in  $\theta$ :*  $\partial_\theta L = 2\lambda\theta - \sum_i (\alpha_i^+ - \alpha_i^-) \phi(x_i) = 0$ , so  $\hat{\theta} = \frac{1}{2\lambda} \sum_i \alpha_i \phi(x_i)$  with  $\alpha_i = \alpha_i^+ - \alpha_i^-$  — the optimum lies in the data span. Substituting, the  $\theta$ -terms become  $\lambda \|\hat{\theta}\|^2 - \sum_i \alpha_i \hat{\theta}^\top \phi(x_i) = -\frac{1}{4\lambda} \sum_{i,j} \alpha_i \alpha_j k(x_i, x_j) = -\frac{1}{4\lambda} \alpha^\top K \alpha$ . *Stationarity in  $\xi$ :*  $\partial_{\xi_i^\pm} L = \frac{1}{n} - \alpha_i^\pm - \gamma_i^\pm = 0$  gives  $\alpha_i^\pm = \frac{1}{n} - \gamma_i^\pm$ , and  $\gamma_i^\pm \geq 0$  forces  $0 \leq \alpha_i^\pm \leq \frac{1}{n}$ , hence  $-\frac{1}{n} \leq \alpha_i \leq \frac{1}{n}$ . Collecting the surviving terms with  $|\alpha_i| = \alpha_i^+ + \alpha_i^-$ ,

$$\tilde{L}(\alpha) = -\frac{1}{4\lambda} \alpha^\top K \alpha - \varepsilon \sum_i |\alpha_i| + \sum_i y_i \alpha_i,$$

and rescaling  $\alpha \leftarrow \alpha/(2\lambda)$  turns the maximisation of  $\tilde{L}$  into the stated minimisation with box constraint  $|\alpha_i| \leq \frac{1}{2\lambda n}$ . Complementary slackness gives sparsity:  $|y_i - \hat{y}(x_i)| < \varepsilon \Rightarrow \alpha_i^+ = \alpha_i^- = 0 \Rightarrow \alpha_i = 0$ , so only support vectors enter  $\hat{y}(x_\star) = \sum_i \alpha_i k(x_i, x_\star)$ .  $\square$

### 14.3 The SVC (soft-margin SVM) dual

#### Plain-language idea

For classification, the  $\varepsilon$ -tube becomes a *margin*: correctly classified points beyond the margin cost nothing (hinge loss). The dual is again a kernel quadratic with box constraints; the support vectors are exactly the “hard” points sitting on or inside the margin.

**Result/Derivation 14.3** (SVC dual). The soft-margin SVM  $\min_{\theta, \xi} \frac{1}{n} \sum_i \xi_i + \lambda \|\theta\|^2$  s.t.  $\xi_i \geq 1 - y_i \theta^\top \phi(x_i)$ ,  $\xi_i \geq 0$ , has  $\theta = \frac{1}{2\lambda} \sum_i \alpha_i y_i \phi(x_i)$  and (after rescaling) dual

$$\min_{\alpha} \frac{1}{2} \alpha^\top (YKY) \alpha - \mathbf{1}^\top \alpha \quad \text{s.t.} \quad 0 \leq \alpha_i \leq \frac{1}{2\lambda n},$$

where  $Y = \text{diag}(y_1, \dots, y_n)$ .

*Proof.* With multipliers  $\alpha_i \geq 0$  (margin) and  $\mu_i \geq 0$  (for  $\xi_i \geq 0$ ),

$$L = \frac{1}{n} \sum_i \xi_i + \lambda \|\theta\|^2 + \sum_i \alpha_i (1 - y_i \theta^\top \phi(x_i) - \xi_i) - \sum_i \mu_i \xi_i.$$

$\partial_\theta L = 2\lambda\theta - \sum_i \alpha_i y_i \phi(x_i) = 0$  gives  $\theta = \frac{1}{2\lambda} \sum_i \alpha_i y_i \phi(x_i)$ .  $\partial_{\xi_i} L = \frac{1}{n} - \alpha_i - \mu_i = 0$  with  $\mu_i \geq 0$  forces  $0 \leq \alpha_i \leq \frac{1}{n}$ . Substituting  $\theta$  and using  $k(x_i, x_j) = \phi(x_i)^\top \phi(x_j)$ , the Lagrangian reduces to  $\sum_i \alpha_i - \frac{1}{4\lambda} \sum_{i,j} \alpha_i \alpha_j y_i y_j k(x_i, x_j)$ . Rescaling  $\alpha \leftarrow \alpha/(2\lambda)$  yields the stated quadratic with  $YKY$  and box constraint  $0 \leq \alpha_i \leq \frac{1}{2\lambda n}$ . The decision function is  $f(x) = \sum_i \alpha_i y_i k(x_i, x)$ ,  $\hat{y}(x) = \text{sign } f(x)$ ; by complementary slackness  $y_i f(x_i) > 1 \Rightarrow \alpha_i = 0$ , so only margin points are support vectors.  $\square$

## 15 Kernel Methods: Gaussian Processes

### 15.1 Construction and existence

#### Plain-language idea

A Gaussian process is a prior over *functions*, defined by saying that the function's values at any finite set of inputs are jointly Gaussian with mean  $\mu(\cdot)$  and covariance given by a kernel. These finite snapshots glue into a genuine random function precisely when the kernel is PSD.

**Theorem 15.1** (Existence via Kolmogorov consistency — stated). If  $\kappa$  is PSD then for every finite  $\{x_1, \dots, x_n\}$  the proposed marginals  $(f(x_1), \dots, f(x_n)) \sim \mathcal{N}(\mu, K)$ ,  $[K]_{ij} = \kappa(x_i, x_j)$ , are mutually consistent, and by the Kolmogorov extension theorem a stochastic process  $f \sim \text{GP}(\mu, \kappa)$  with these marginals exists. PSD-ness is also necessary for each  $K$  to be a valid covariance.

A linear model  $f(x) = \langle \phi(x), \theta \rangle$  with  $\theta \sim \mathcal{N}(\mu_\theta, I)$  is a GP:  $f = \Phi\theta \sim \mathcal{N}(\Phi\mu_\theta, \Phi\Phi^\top)$ , so  $\mu(x) = \langle \phi(x), \mu_\theta \rangle$  and  $\kappa(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$ .

### 15.2 The marginal distribution of the observations

**Result/Derivation 15.2** (Distribution of  $y$ ). With  $f \sim \mathcal{N}(\mu, K)$  and independent noise  $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ , the observations  $y = f + \varepsilon$  satisfy  $y \sim \mathcal{N}(\mu, K + \sigma^2 I)$ .

*Proof.* The sum of independent Gaussians is Gaussian with added means and added covariances:  $\mu_a + \mu_b = \mu + 0 = \mu$  and  $\Sigma_a + \Sigma_b = K + \sigma^2 I$ .  $\square$

### 15.3 GP regression posterior by Gaussian conditioning

#### Plain-language idea

Because training values, the test value, and noise are all jointly Gaussian, predicting the test value is just *conditioning* a big Gaussian on the observed coordinates. The standard conditioning formula hands us the posterior mean (prior mean plus a data correction) and posterior variance (prior variance minus what the data explained).

#### Approach

Build the joint distribution of  $(y, f_\star)$  from the GP property (function values are jointly Gaussian) and affine closure (adding independent noise), read off its mean and block covariance, then apply the Gaussian conditioning identity.

**Result/Derivation 15.3** (GP posterior). Let  $\mu_\star = \mu(x_\star)$ ,  $[k_\star]_i = \kappa(x_i, x_\star)$ ,  $\kappa_{\star\star} = \kappa(x_\star, x_\star)$ . The joint law is

$$\begin{bmatrix} y \\ f_\star \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu \\ \mu_\star \end{bmatrix}, \begin{bmatrix} K + \sigma^2 I & k_\star \\ k_\star^\top & \kappa_{\star\star} \end{bmatrix}\right),$$

and conditioning gives

$$f_\star | y \sim \mathcal{N}\left(\underbrace{\mu_\star + k_\star^\top (K + \sigma^2 I)^{-1} (y - \mu)}_{\mu_n(x_\star)}, \underbrace{\kappa_{\star\star} - k_\star^\top (K + \sigma^2 I)^{-1} k_\star}_{\sigma_n^2(x_\star)}\right).$$

*Proof.*  $f$  and  $f_\star$  are jointly Gaussian by the GP property; adding independent  $\varepsilon$  to  $f$  is an affine map of the Gaussian vector  $(f, f_\star, \varepsilon)$ , so  $(y, f_\star)$  is jointly Gaussian. Its mean is  $(\mu, \mu_\star)$ . The blocks are  $\text{Cov}(y, y) = K + \sigma^2 I$ ,  $\text{Cov}(y, f_\star) = \text{Cov}(f + \varepsilon, f_\star) = \text{Cov}(f, f_\star) = k_\star$  (noise



is independent of  $f_\star$ ), and  $\text{Cov}(f_\star, f_\star) = \kappa_{\star\star}$ . Applying the Gaussian conditioning rule: if  $\begin{bmatrix} a \\ b \end{bmatrix} \sim \mathcal{N}(\begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \begin{bmatrix} A & C \\ C^\top & B \end{bmatrix})$  then  $b \mid a \sim \mathcal{N}(\mu_b + C^\top A^{-1}(a - \mu_a), B - C^\top A^{-1}C)$ , with  $a = y$ ,  $b = f_\star$ ,  $A = K + \sigma^2 I$ ,  $C = k_\star$ ,  $B = \kappa_{\star\star}$ , gives the stated  $\mu_n(x_\star)$  and  $\sigma_n^2(x_\star)$ . The predictive distribution for a noisy observation adds  $\sigma^2$ :  $y_\star \mid y \sim \mathcal{N}(\mu_n(x_\star), \sigma_n^2(x_\star) + \sigma^2)$ .  $\square$

## 15.4 Posterior variance is monotone decreasing in the data

### Plain-language idea

More observations can only ever *reduce* (never increase) the GP's uncertainty at any point: extra data places more constraints on which functions are plausible, so the explained-away part of the prior variance can only grow.

**Theorem 15.4** (Monotone uncertainty). If  $D_1 \subseteq D_2$  then for every test point  $x$ ,  $\sigma_{D_2}^2(x) \leq \sigma_{D_1}^2(x)$ .

*Proof intuition (kernel viewpoint, as in the lecture).* The posterior variance is  $\sigma^2(x) = \kappa(x, x) - k^\top (K + \sigma^2 I)^{-1} k$  — prior uncertainty minus the component explained by data. Enlarging the dataset appends rows/columns to  $K$  and entries to  $k$ ; the resulting quadratic form  $k^\top (K + \sigma^2 I)^{-1} k$  can only increase (an additional observation is an extra orthogonal direction of explanation it cannot remove). Hence the remaining unexplained variance  $\sigma^2(x)$  cannot increase at any  $x$ . A fully rigorous proof uses the Schur-complement/positive-semidefinite ordering of the conditional covariances.  $\square$

The GP log marginal likelihood  $\log p(y) = -\frac{1}{2}(y - \mu)^\top (K + \sigma^2 I)^{-1} (y - \mu) - \frac{1}{2} \log |K + \sigma^2 I| - \frac{n}{2} \log(2\pi)$  follows directly from  $y \sim \mathcal{N}(\mu, K + \sigma^2 I)$ , and is maximised over kernel hyperparameters for model selection (data-fit term vs. log-determinant complexity penalty).

## Closing note

This compendium covers every theorem-with-proof and every substantive derivation in the lecture decks. A handful of results are stated by the course without proof and are flagged as such here: Stone's universal-consistency theorem, the Moore–Aronszajn theorem, the Kolmogorov consistency/existence of a GP, and the rigorous monotonicity of GP posterior variance (given with the lecturer's kernel-viewpoint intuition). The conceptual *Introduction* and the bulk of the *Modern View* of model complexity contain no formal proofs and are summarised rather than reproduced.